Lecture-4
Fluid Statics II: Forces on Surfaces

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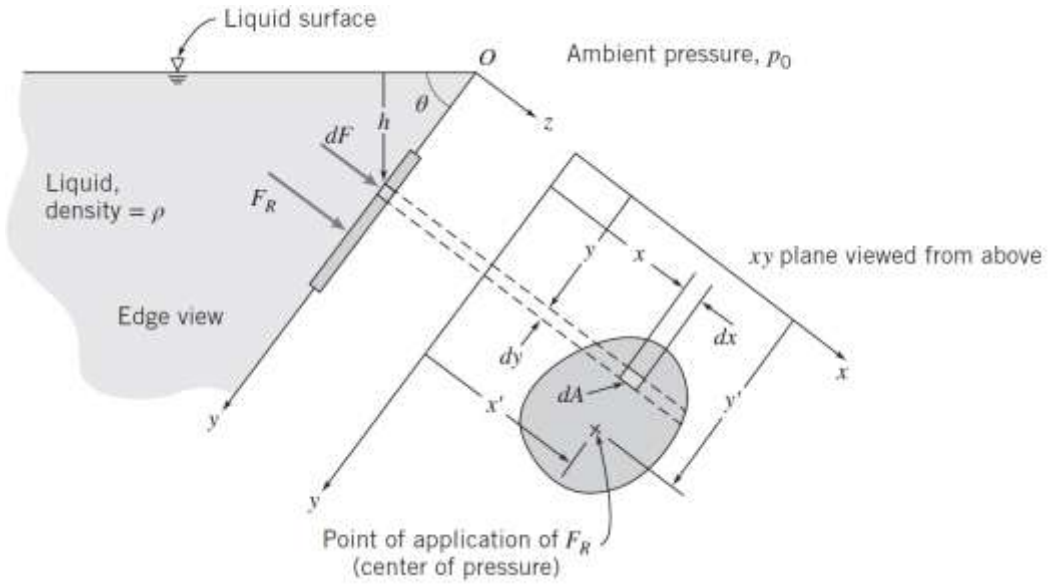
Learning Objectives

By the end of this lecture, you should be able to

1. Explain **why a force acts** on a surface
2. Give analytical expressions for the **magnitude and direction** of the force
3. Understand the similarities (differences) between the forces experienced by **plane versus curved** surfaces

Force on a plane surface

- Since fluid pressure exerts a normal force per unit area, **a finite surface area** within the fluid domain is subject to **a distributed force**.
- To specify the force completely, we need to determine the **magnitude, direction, and line of action of the force**



$$F_R = \int_A p \, dA$$

$$F_R = p_c A$$

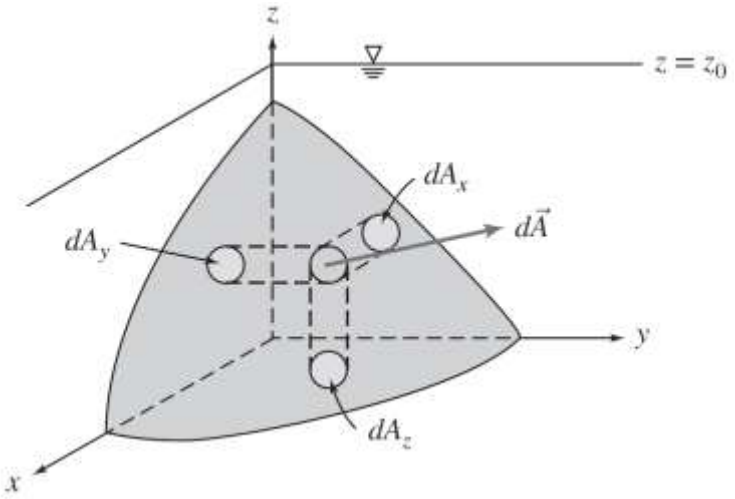
$$y' = y_c + \frac{\rho g \sin\theta I_{\hat{x}\hat{x}}}{F_R}$$

$$x' = x_c + \frac{\rho g \sin\theta I_{\hat{x}\hat{y}}}{F_R}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Force on a curved surface

- The area element for a curved surface **has a different direction** at every point on the surface, making its analysis slightly different.



$$\vec{F}_R = -\int_A p\, d\vec{A}$$

$$F_{R_l} = \int_{A_l} p\, dA_l$$

- The **horizontal force** and its location are the same as for an **imaginary vertical plane surface of the same projected area.**
- The **net vertical force** will be equal to the **weight of the fluid** directly above the surface**.**

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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